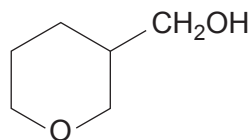
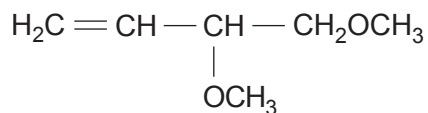


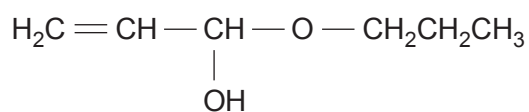
3. Four isomers of formula $C_6H_{12}O_2$ are given below.



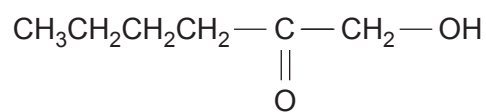
compound **D**



compound **E**



compound **F**



compound **G**

Deduce which isomer has an infrared absorption spectrum showing absorptions at 1620 cm^{-1} and at $3200\text{--}3500\text{ cm}^{-1}$ but **not** at $1650\text{--}1750\text{ cm}^{-1}$. Give reasons for your answer. [2]

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4. Give the formula of the triphenylmethyl radical. [1]

